



Perturbation-specific transcriptional mapping for unbiased target elucidation of antibiotics

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The rising prevalence of antibiotic resistance threatens human health. While more sophisticated strategies for antibiotic discovery are being developed, target elucidation of new chemical entities remains challenging. In the postgenomic era, expression profiling can play an important role in mechanism-of-action (MOA) prediction by reporting on the cellular response to perturbation. However, the broad application of transcriptomics has yet to fulfill its promise of transforming target elucidation due to challenges in identifying the most relevant, direct responses to target inhibition. We developed an unbiased strategy for MOA prediction, called perturbation-specific transcriptional mapping (PerSpecTM), in which large-throughput expression profiling of wild-type or hypomorphic mutants, depleted for essential targets, enables a computational strategy to address this challenge. We applied PerSpecTM to perform reference-based MOA prediction based on the principle that similar perturbations, whether chemical or genetic, will elicit similar transcriptional responses. Using this approach, we elucidated the MOAs of three molecules with activity against *Pseudomonas aeruginosa* by comparing their expression profiles to those of a reference set of antimicrobial compounds with known MOAs. We also show that transcriptional responses to small-molecule inhibition resemble those resulting from genetic depletion of essential targets by clustered regularly interspaced short palindromic repeats interference (CRISPRi) by PerSpecTM, demonstrating proof of concept that correlations between expression profiles of small-molecule and genetic perturbations can facilitate MOA prediction when no chemical entities exist to serve as a reference. Empowered by PerSpecTM, this work lays the foundation for an unbiased, readily scalable, systematic reference-based strategy for MOA elucidation that could transform antibiotic discovery efforts.

antibiotics | RNAseq | gene expression | mechanism-of-action | transcriptomics

Pseudomonas aeruginosa and other gram-negative bacteria pose a major threat to human health due to the growing prevalence of antibiotic resistance in healthcare settings. Surging multidrug-resistant bacterial infections threaten to outpace our ability to develop novel antibiotics (1). Antibiotic discovery campaigns have failed to deliver new antibiotic classes, including extensive efforts in the postgenomic era. As recently reviewed, new strategies are being developed to address the looming antibiotic shortage, including novel approaches to mine the natural products space (2–4), to apply AI (5, 6), and to leverage genetic engineering of strains in a chemical systems biology approach (7, 8). Most of these novel approaches differ from standard in vitro, target based strategies in which the target is known by design, as they prioritize the identification of new molecules with whole-cell antibacterial activity. While accelerating the discovery of new chemical entities, these modern approaches highlight the downstream challenge of mechanism-of-action (MOA) determination and the need for better, more systematic methods to tackle this problem to enable candidate development. Currently, extensive biochemical and genetic efforts are required for target and MOA elucidation, which impose a major bottleneck in antibiotic discovery and development pipelines (9). Indeed, early MOA assignment for antibacterial small-molecule candidates would provide mechanistic and biological insights to guide their prioritization, medicinal chemistry optimization, and eventual hit-to-lead transition.

Transcriptional profiling has long been used to explore the MOA of antimicrobial agents, first described in yeast using microarrays to delineate the MOA of the inhibitor dyclonine (10), and later performed in a wide array of bacteria, including *Bacillus subtilis*, *Mycobacterium tuberculosis*, *Staphylococcus aureus*, *Escherichia coli*, and *P. aeruginosa* (11–17). Work in *B. subtilis* generated the first antimicrobial reference set based on gene expression using microarrays to characterize the transcriptional responses to 37 antibiotics (16). More recently, work in *E. coli* also included a reference set of 37 antibiotics, from

Significance

New antibiotics are critically needed in the face of increasing antibiotic resistance. However, mechanism-of-action (MOA) elucidation remains challenging and imposes a major bottleneck in antibiotic discovery and development. Building on the principle that molecules with similar MOAs elicit similar transcriptional responses, we have developed a highly scalable strategy for MOA prediction in the important bacterial pathogen *Pseudomonas aeruginosa* based on correlations between the expression profiles of molecules having an unknown MOA and the expression profiles resulting from known perturbations, either small-molecule inhibition by known antibiotics or transcriptional repression of essential targets by CRISPRi. By rapidly assigning MOAs to three molecules with antipseudomonal activity, we provide proof of concept for a rapid, comprehensive, systematic, reference-based approach to MOA prediction with the potential to transform antibiotic discovery efforts.

The authors declare no competing interest.

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which the authors chose gene sets based on differential gene expression analysis of compound treatment relative to negative control, and used correlation coefficients to predict MOAs (12). Then, Espinoza et al. furthered this work by developing a machine learning approach for transcriptomics-based MOA prediction in *E. coli*, which involved a reference set of 41 antibiotics and a feature selection algorithm called *Clairvoyance* (13).

Despite these advances, transcriptomics-based MOA prediction remains limited by the challenge of extracting meaningful signals that distinguish the relevant gene responses to target inhibition from less specific transcriptional changes (18, 19). Ideally, transcriptomics-based MOA prediction should be rooted in gene expression responses specific to the given chemical or genetic perturbation. However, during perturbation, bacteria also change the expression of a plethora of background genes, which may be far downstream from the perturbed pathway or unrelated altogether, which obfuscate the perturbation-specific signals. For example, nonspecific gene responses, relating to cellular stress or impending cell death, can be shared across antimicrobial compounds with disparate MOAs. Meanwhile, differences in measured expression programs can also be related to growth phase differences between treated and untreated cells as they grow or divide at different rates, rather than a direct effect of antimicrobial activity. To manage these effects, treatments with sublethal concentrations of compound are often used to minimize the growth arrest or death of treated populations and match growth rates of treated and untreated samples; however, these low treatment concentrations often yield weak transcriptional responses, thereby obscuring informative, relevant changes that occur with compound exposure (11–17).

To address this challenge, we developed a systematic method called perturbation-specific transcriptional mapping (PerSpecTM) that combines experimental and analytic methods that leverage its large-throughput scale to accentuate changes in gene expression that are specific to and a direct consequence of target inhibition. By using a diverse set of perturbed transcriptomes for normalization, an approach adapted from large-scale gene profiling in human cancer cell lines (20), PerSpecTM emphasizes perturbation-specific transcriptional signatures while deemphasizing common, nonspecific responses. It then assigns target and MOA in a reference-based manner by correlating the pattern of changes with the transcriptional responses of a reference set, thus avoiding biases that might occur by assigning MOA based on the response of specific, predetermined genes.

We applied PerSpecTM to the important bacterial pathogen *P. aeruginosa*, building two complementary reference sets: 1) an antimicrobial reference set comprising 37 antimicrobial compounds and 2) a genetic depletion reference set comprising 14 strains, each exhibiting tunable transcriptional repression of a different essential gene target using CRISPRi. Querying against the antimicrobial reference set, we rapidly elucidated the targets and MOAs of three molecules with antipseudomonal activity, while demonstrating the value of using genetically engineered, hypersensitized strains depleted for essential targets to yield more robust transcriptional responses to perturbation. Because of the limited number of chemical entities with known unique targets, and thereby MOAs, that can serve as an antimicrobial reference, we applied PerSpecTM to the transcriptional repression reference set to demonstrate proof of concept that genetically repressing target expression can phenocopy its chemical inhibition, thus affording target and MOA prediction when no chemical entities exist to serve as a reference. Empowered by PerSpecTM and mutant strains depleted in essential targets, this work enables an unbiased and highly scalable reference-based strategy for target and MOA elucidation to assist antimicrobial discovery efforts in

the face of the looming shortage of antibiotics in healthcare systems worldwide.

Results

Developing PerSpecTM. We set out to generate in large-throughput, an antimicrobial reference set of chemical treatments of the *P. aeruginosa* laboratory strain UCBPP-PA14 (abbreviated hereafter as PA14) and related mutant strains. To enable such scale, we worked to determine the optimal assay conditions in 384-well format and analysis strategy in a manner that would not require tailoring conditions for each individual compound.

All expression profiling experiments were performed in 384-well format with 60 μ L per well of bacterial culture at an inoculum of $\sim 1 \times 10^8$ colony-forming unit (CFU) per milliliter (mL), used to ensure that sufficient cells were available to guarantee the generation of a high-quality ribonucleic acid sequencing (RNAseq) library. Each compound was dosed in triplicate at threefold the minimum-inhibitory concentrations (MICs), with the MIC determined by conventional broth microdilution at a much lower inoculum of $\sim 5 \times 10^5$ CFU/mL (21). Plates were arrayed with a diverse set of active compounds across multiple MOA classes, as well as dimethylsulfoxide (DMSO) and water controls. Resulting gene expression data for each gene of each sample were standardized (transformed to z-scores) with respect to all samples in the same batch, where each batch is defined as samples on the same 384-well plate sharing the same strain and time point. The z-score provides a measure of how unusually high or low the expression of a given gene is after a specific chemical (or genetic) perturbation relative to its expression across the entire population of samples within a batch. Provided the batch consists of a diverse set of active antimicrobial agents (or genetic perturbations), z-scores highlight transcriptional changes that are unique to a given perturbation, and deemphasize common gene responses across the population, such as those relating nonspecifically to growth, cellular stress, or impending cell death. Thus, the relevant signals are accentuated in an unbiased manner, without needing to subset genes based on computational approaches or a priori knowledge of expected gene responses. The ability to calculate z-scores across a variety of treatments, rather than just assessing differential expression between a particular sample and negative control samples, was made possible by the large-throughput scale of PerSpecTM. To then assess overall strength of the transcriptional signals, differential gene expression was orthogonally assessed using DESeq2 (22). A “transcriptionally weak signal” was ascribed to a treatment condition when no gene was differentially expressed (compared to vehicle control) with an adjusted *P*-value below 10^{-20} [i.e., no gene had a $-\log_{10}(p_{\text{adj}}) > 20$], which represents the minimum adjusted *P*-value across all differentially expressed genes of PA14 in water versus PA14 in DMSO. Therefore, all differentially expressed genes in transcriptionally weak samples were less statistically significant than the most significant differentially expressed genes across vehicle controls. Although weak signals may reflect meaningful biological changes, we wanted to only include robust signals in the antimicrobial reference set to facilitate reliable target predictions. Similarities between perturbations were then calculated using the Pearson correlation coefficients between transcriptional signatures of two conditions, where each gene’s expression is represented as a z-score, with replicates having been combined using a weighted average (*Materials and Methods*).

We first set out to determine the optimal incubation time and compound concentration yielding robust transcriptional responses across many different compound treatments, without individual compound optimization. We profiled a set of 21 antimicrobial compounds at 30, 60, 90, or 120 min, depicted in bold in *SI Appendix, Table S1*, which collectively targeted four distinct

cellular mechanisms: inhibitors of DNA synthesis, protein synthesis, cell wall (defined as the structural elements of the cell envelope, excluding the cell membranes), and membrane integrity. After calculating z-scores for each gene and determining the Pearson distances [$\sqrt{1 - \text{Pearson Correlation}}$] between adjacent time points over the 120-min time trial (*SI Appendix, Fig. S1A*), we found that the transcriptional responses to treatment by inhibitors of DNA synthesis, protein synthesis, and membrane integrity converged on transcriptional endpoints by 90 min within each MOA. The transcriptional responses to treatment with cell wall synthesis inhibitors were more heterogeneous as reflected in their larger Pearson distances, which never fully converged by 120 min.

We wondered whether the lack of transcriptional convergence of cell wall inhibitors by 120 min related to the inoculum effect, a phenomenon observed in β -lactams causing loss of activity at high bacterial densities (23). We measured the MICs of compounds at higher inoculum (i.e., 1×10^8 CFU/mL used in RNAseq experiments) compared to the lower convention of $\sim 5 \times 10^5$ CFU/mL (21). *SI Appendix, Table S1* denotes the antimicrobial compounds exhibiting strong inoculum effects ($\geq$ fourfold shift in MIC at the high bacterial density) with an asterisk, which comprise all of the cell wall synthesis inhibitors, except imipenem, due to upregulation of the beta lactamase AmpC at high bacterial density (24). Smaller inoculum effects were also observed for tobramycin and several antimicrobial peptides, which have been previously reported through various proposed mechanisms, including growth phase changes, stress responses, outer membrane vesicle production, and inactivating enzymes (25–27). Interestingly, the Pearson distances for imipenem across the time trial also converge by 90 min, comparable to inhibitors of protein synthesis, DNA synthesis, and membrane integrity (*SI Appendix, Fig. S1A*). Taken together, these data suggested that for antimicrobial compounds which are active at the experimental inoculum, an incubation time of 90 min yielded a stable transcriptional endpoint.

After establishing an optimal 90-min incubation time, we sought to determine the optimal antimicrobial concentration that elicits a strong transcriptional response. Here, we used an expanded set of 39 antimicrobial compounds, shown in *SI Appendix, Table S1*, spanning the same four MOAs but through inhibition of more diverse targets. We added several compounds with known antipseudomonal activity, but not yet approved for human use. For example, we included cell wall-related compounds, such as inhibitors of LpxC, LptD, BamA, and MreB that disrupt lipopolysaccharide (LPS) biosynthesis, LPS transport, outer membrane protein assembly, and cell shape, respectively. We also included the nonpseudomonal antimicrobial compound nitrofurantoin, which works through a complex mechanism involving reactive oxygen species that interfere nonspecifically with DNA synthesis, as well as protein and cell wall metabolism (28). We also added 1) the antibacterial agent fosmidomycin, which inhibits Dxr to block isoprenoid synthesis (29); 2) the Hsp90 inhibitor radicicol, an antifungal agent that likely interferes with gyrase-related proteins with a conserved Bergerat fold (30–32); 3) the antistaphylococcal agent fusidic acid, an inhibitor of the elongation factor FusA1 that blocks translation (33); 4) the antibiotic rifampicin, which blocks transcription through the inhibition of the DNA-dependent RNA polymerase RpoB (34); 5) the fatty acid synthesis inhibitor thiolactomycin; and 6) the membrane pore former LL-37 (35). Finally, we included several anticancer agents, such as cisplatin, doxorubicin, and hydroxyurea, which exhibit antimicrobial activity by inhibiting DNA synthesis through DNA alkylation, DNA depletion, and DNA intercalation, respectively (36–42).

Compounds were tested at 0.5-fold, twofold, and fourfold their adjusted MICs as determined in broth microdilution experiments at high inoculum, thereby accounting for any inoculum effects. In some

cases, the highest possible concentrations were limited by solubility. These data confirmed that all compounds dosed above their MICs demonstrated robust transcriptional responses (*SI Appendix, Fig. S1B*). In contrast, antimicrobials that were inactive (nonlethal) at the applied concentrations yielded variable transcriptional responses. In some cases, such as for DNA synthesis inhibitors and membrane disruptors, robust gene expression signals were generated even under subinhibitory concentrations, suggesting that bacteria are primed to detect nonlethal perturbations of certain cellular processes. In the case of some of the β -lactam antibiotics, such as piperacillin, ceftazidime, and aztreonam, robust transcriptional responses could not be elicited due to compound insolubility at concentrations necessary to overcome the inoculum effect (*SI Appendix, Fig. S1C*). In fact, because aztreonam treatment yielded a transcriptionally weak response across all treatment conditions, it was removed from the reference set moving forward. Taken together, these results demonstrate that antimicrobial activity at the high experimental inoculum (i.e., 1×10^8 CFU/mL) was necessary to ensure a robust transcriptional response to all antibiotics in this large-throughput assay format, even while some compounds are still able to generate robust signals despite sublethal dosing. We thus settled on the general use of 2 to 4 $\times$ the MIC measured at a high inoculum for compound treatments.

Transcriptional Responses to Antimicrobial Compounds with Similar MOAs Are Correlated.

As expected, our large-throughput gene expression dataset confirmed that lethal antimicrobial treatments elicited transcriptional responses that correlated by MOA. The Pearson correlations of z-scores of pairwise combinations of 90-min treatments, at the highest dose for each compound, are shown as a similarity matrix in Fig. 1A. Compounds with limited antipseudomonal activity, comprising doxorubicin, radicicol, hydroxyurea, piperacillin, and rifampicin, had positive correlation values with vehicle treatment. Of the remaining antimicrobial compounds, Pearson correlations were strongest between compounds sharing the same cellular mechanism. Not only did compounds correlate by MOAs, but stronger correlations were observed within antibiotic subclasses. For example, based on the strength of their correlations, protein synthesis inhibitors grouped into macrolides (clarithromycin and erythromycin), tetracyclines (tetracycline, minocycline, and doxycycline), and aminoglycosides (amikacin and gentamicin). Likewise, fluoroquinolones (ciprofloxacin and levofloxacin) formed a clear subgroup within the larger class of DNA synthesis inhibitors. Finally, the transcriptional responses to LptD inhibitors (POL-7001 and POL-7080) and LpxC inhibitors (PF-5081090 and PF-04753299), which block LPS synthesis and transport to the cell surface, respectively, tightly correlated within the group of membrane integrity inhibitors. Meanwhile, despite the resolution obtained for some subclasses, inhibitors of membrane integrity and cell wall synthesis elicited transcriptional responses that cross-correlated, pointing to a shared cellular response to defects in membrane disruption and cell wall synthesis.

Finally, to verify that the transcriptional responses to compound treatment corresponded with known biology, we determined the most up-regulated and down-regulated genes that emerged from calculating z-scores in response to three of the most well-studied antibiotic classes: aminoglycosides, fluoroquinolones, and polymyxins. *SI Appendix, Table S2* shows the top 25 up-regulated and down-regulated genes with known annotations in PA14 in response to treatment with aminoglycosides, fluoroquinolone, and polymyxins. Indeed, previously reported gene expression signatures were readily apparent for each antibiotic class, providing a sanity check that the PerSpecTM approach indeed highlights changes in gene expression directly related to each specific chemical perturbation.

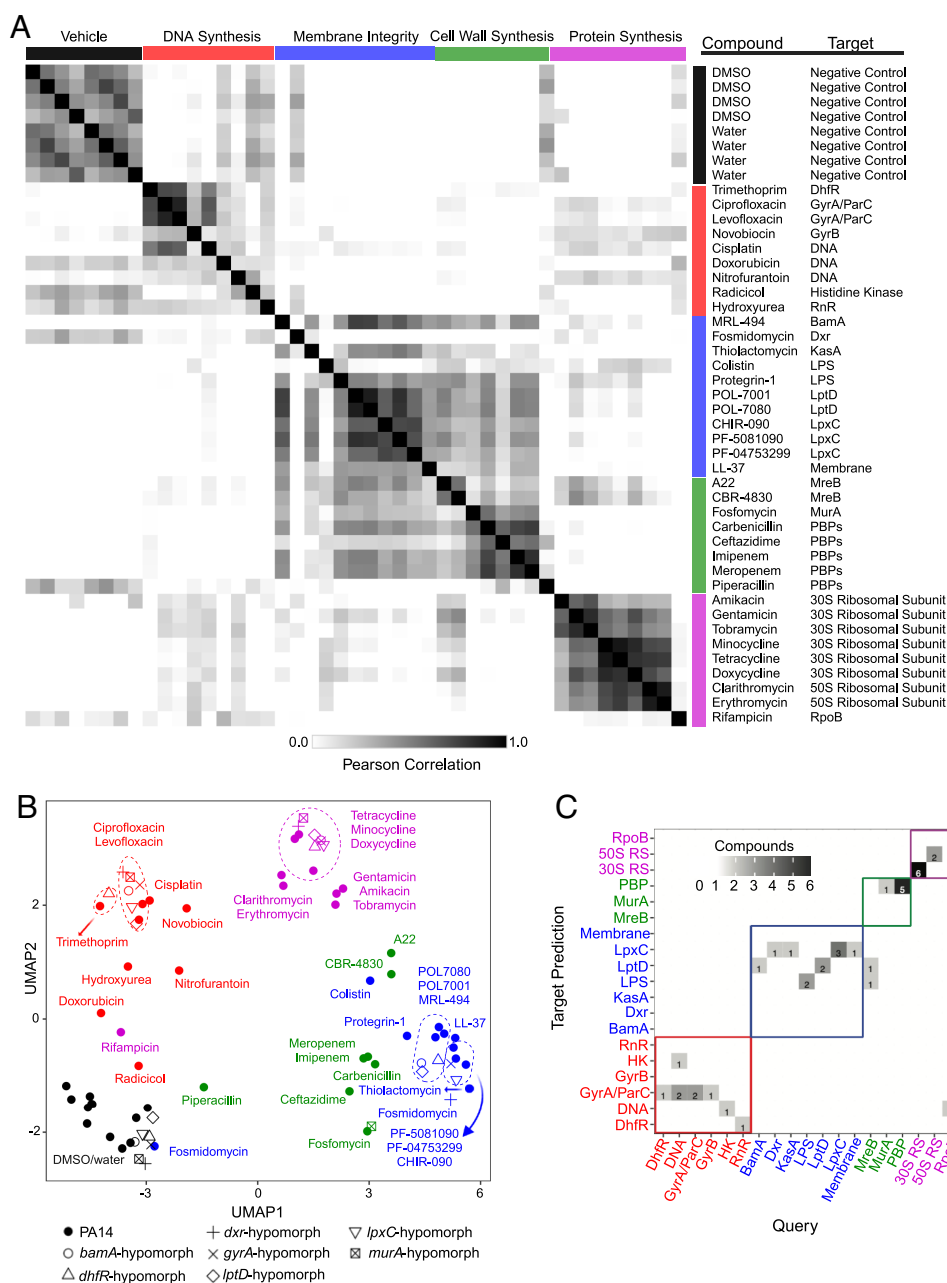


Fig. 1. Correlating antimicrobial gene expression profiles of the antimicrobial reference set. (A) Heat map of Pearson correlations of the replicate-collapsed, z-score expression profiles of PA14 treated with all 37 compounds in the antimicrobial reference set at the highest concentration at 90 min. Rows and columns are organized and color-coded by MOA. All correlation values below zero were set to zero for visualization purposes. (B) UMAP visualization of all compound treatments (at the highest dose) of wild-type PA14 and seven engineered hypomorphs at 90 min. Colors correspond to antibiotic MOA classes: DNA synthesis (red), membrane integrity (blue), cell wall synthesis (green), and protein synthesis (magenta). Closed circles indicate wild-type PA14 treatments, whereas all other markers indicate treatment of a hypomorph strain, as indicated. Antimicrobial compounds with robust activity at the experimental inoculum group by MOA, agnostic of the genetic strain. Protein synthesis inhibitors form subgroups that separate tetracyclines, macrolides, and aminoglycosides. Except for novobiocin, compounds that are not active against PA14, such as fosmidomycin, piperacillin, rifampicin, doxorubicin, and hydroxyurea, appear close to DMSO vehicle. Treatments that were deemed transcriptionally weak based on DESeq2 analysis were not included in the visualization. (C) Confusion matrix showing the number of compounds in each query target category that were predicted to have indicated targets in the leave-one-out cross-validation (LOOCV), using the method of calculating the target correlation prediction score (r_{predict}). Compounds with correctly predicted target fall along the diagonal, while those with correctly predicted MOA fall within the colored boxes. Note that query target categories that only contain one compound (the column sums to one) could not be predicted correctly, due to the LOO approach.

Transcriptional Responses to Treatment of Hypersusceptible Strains Correlate with Wild-Type Treatments.

The requirement that compounds have potent wild-type activity to elicit a reliable, robust, informative transcriptional response poses a barrier to the early integration of transcriptomics-based MOA prediction for newly discovered compounds which might not yet have highly potent antimicrobial activity. To address this concern, we sought to determine whether mutant strains that are hypersusceptible to

compounds could be used in place of wild-type strains to obtain robust transcriptional signals for compounds with limited wild-type activity. While an efflux pump mutant could serve as a hypersusceptible strain to some compounds, not all compounds have increased sensitivity in such a strain. We thus selected an alternative strategy involving the engineering of hypomorph mutants that are hypersensitized to inhibitors by depleting their cognate gene products (8).

To test this concept, we sought to create hypomorphs and compare their transcriptional responses to antimicrobial compounds with those of wild-type bacteria. We genetically engineered seven hypomorphic strains using a regulated promoter replacement strategy to deplete essential gene targets corresponding to a set of known antimicrobial agents (*SI Appendix, Additional Methods* for details of strain construction). As summarized in *SI Appendix, Table S3*, the panel of hypomorphic strains compromised gene target depletions of the LPS transporter protein LptD, the isoprenoid synthesis protein Dxr, the peptidoglycan synthesis protein MurA, DNA gyrase GyrA, the LPS biosynthesis enzyme LpxC, the dihydrofolate reductase Dhfr involved in folate biosynthesis, and the outer membrane protein assembly protein BamA. The growth phenotypes were unchanged for most engineered hypomorphs, but reduced in the case of *gyrA*, *dxr*, and *lptD* hypomorphs (*SI Appendix, Fig. S2*). We then measured the MICs of compounds targeting each depleted gene product: POL7080 against LptD (43), fosmidomycin against Dxr (29), fosfomycin against MurA (44), ciprofloxacin against GyrA (45), PF-04753299 against LpxC (46), trimethoprim against Dhfr (47), and MRL-494 against BamA (48). Except for the *lptD*-hypomorph against doxycycline, the hypomorphs did not show appreciable hypersensitivity to inhibitors whose MOA were unrelated to the specific target depleted in each strain. By comparison, the *lptD*, *murA*, *dxr*, and *bamA* hypomorphs were 4 to 20-fold more susceptible to their cognate inhibitors, whereas the *gyrA*, *lpxC*, and *dhfr* hypomorphs showed no significant shifts in MIC relative to the wild-type (*SI Appendix, Table S3*). Of note, the *gyrA* and *dhfr* hypomorphs also showed no reduced transcription of their respective genes (*SI Appendix, Table S3*) consistent with the hypothesis that their respective replaced promoters did not achieve the desired goal of decreasing gene expression and thus target depletion. We did nevertheless include them in all subsequent studies to maximize the statistical power of the studies, treating them like the wild-type.

We measured the gene expression profile of each hypomorph treated with vehicle, one on-target compound (i.e., compound whose activity is due to inhibition of the corresponding target), and two off-target compounds (i.e., compound whose activity is not related to the corresponding target). Hypomorphic strains were incubated with compounds at twofold their MICs (determined at high inoculum against each hypomorph), or vehicle. Z-scores were calculated across each hypomorph, rather than across all hypomorphs, to dampen background transcriptional changes related to strain-specific repression in gene expression. Fig. 1*B* shows a combined uniform manifold approximation and projection (UMAP) showing wild-type and all mutants with their corresponding treatments. The changes in gene expression of hypomorphs to chemical perturbation correlated with wild-type treatment responses. For example, treatment of the *gyrA*-hypomorph with its on-target inhibitor, ciprofloxacin, grouped with PA14 treatment with ciprofloxacin, as was the case for *lpxC*-hypomorph treated with PF-04753299, *bamA*-hypomorph treated with MRL-494, *dhfr*-hypomorph treated with trimethoprim, *murA*-hypomorph treated with fosfomycin, and *lptD*-hypomorph treated with POL7080. Meanwhile, grouping of mutant and wild-type responses was agnostic of whether a compound's molecular target related to the underlying genetic alteration in a strain, as exemplified by the treatment response of the *gyrA*-hypomorph to doxycycline, a protein synthesis inhibitor unrelated to GyrA, grouping together with PA14 in response to doxycycline.

The correlation between (separately standardized) responses of hypersusceptible mutants and the wild-type enabled the characterization of expression responses of compounds without wild-type activity. Indeed, the transcriptional response of the *dxr*-hypomorph to fosmidomycin provides proof of concept for the utility of using genetically depleted strains to facilitate MOA prediction. Fosmidomycin

inhibits Dxr, an essential reductoisomerase in the 2-C-methyl-D-erythritol 4-phosphate pathway of isoprenoid precursor biosynthesis (49). While fosmidomycin exhibited an MIC of 12.5 μ M at low inoculum, it showed no activity against PA14 at the high inoculum used for expression profiling experiments. In contrast, fosmidomycin caused a significant growth delay of the *dxr*-hypomorph at high inoculum, causing 50% growth inhibition at about 25 μ M (*SI Appendix, Fig. S3*). While the expression response of wild-type PA14 to fosmidomycin treatment grouped near vehicle controls, the response of the *dxr*-hypomorph grouped with inhibitors of membrane integrity, accurately reflecting the cellular mechanism of fosmidomycin (Fig. 1*B*).

MOA and Target Prediction Using the Antimicrobial Reference Set.

The antimicrobial reference set was curated to include robust transcriptional profiles of all aforementioned treatments across various time points, doses, and hypomorphic strains. Treatment conditions eliciting weak transcriptional signals were excluded, which fully excluded aztreonam from the reference set, as described earlier. In addition, any treatment condition having a z-score transcriptional profile that correlated more highly to negative control samples than to any other condition in the dataset, including other treatment conditions of the same compound, was also removed. This resulted in the exclusion of fusidic acid from the reference set. In total, 37 compounds were retained in the antimicrobial reference set.

We next set out to evaluate the ability of PerSpecTM to perform reference-based target and MOA predictions for novel chemical entities using an antimicrobial reference set, with target referring to the biomolecule engaged by a given compound, and MOA referring to the cellular function disrupted by target engagement. We hypothesized that the predicted target for a query compound could be determined by the highest overall Pearson correlation between the query compound and all other treatments within the reference set, similar to methods described previously (12). In practice, to arrive at a single target prediction for a given compound, in cases where there are multiple treatment conditions (e.g., multiple doses or time points), we first determined the maximal Pearson correlation (r_{max}) between each query treatment and every individual treatment within each reference target category. We removed query treatments in which the highest r_{max} values derived from negative control samples. Then, we averaged the r_{max} values from all remaining query treatments for a given compound to define a target correlation score, $\bar{r}_{max}$. In the case where there is only one treatment condition for a query compound, $\bar{r}_{max} = r_{max}$. The final target prediction was assigned based on the highest $\bar{r}_{max}$ across all target categories in the reference set, which we termed the target correlation prediction score, $r_{predict}$. In the case where there is only one treatment condition for a query compound, $r_{predict}$ is the equivalent to the highest overall Pearson correlation between the query compound and all reference treatments. The predicted MOA of the query compound was then assigned as the known cellular pathway associated with the predicted target.

To evaluate the performance of PerSpecTM, we performed a leave-one-out cross-validation (LOOCV) (Fig. 1*C*), wherein the expression profiles for each individual antimicrobial compound in the reference set were queried against the expression profiles of all other compounds in the reference set. The correlation analysis was carried out as described above, correctly assigning MOAs for 34 of the 37 antimicrobial compounds (92%), and correctly assigning targets for 22 of the compounds (60%). Of note, 10 compounds (27%) were the only representatives of their respective target categories, and thus LOOCV had no chance of correct target prediction. Removing those ten singleton compounds from the query, the correct MOAs

were assigned for 25 out of 27 compounds (94%), and correct targets were predicted for 22 of the compounds (81%).

Next, to assess confidence in each MOA and target prediction, we obtained positive predictive values (PPVs) for both MOA and target assignments, respectively. The PPV is the percent of compounds in the LOOCV whose prediction was correct for r_{predict} values above a given threshold. Of note, the distributions of r_{predict} are different for different MOAs, and thus may merit different levels of confidence depending on the predicted MOA (SI Appendix, Fig. S4). However, due to the limited number of compounds per category, the PPVs were calculated across the entire dataset for both MOA and target. For each target and MOA prediction in the subsequent sections, we report the respective PPVs that are associated with the final prediction score, r_{predict} .

Finally, given PerSpecTM's ability to highlight transcriptional changes unique to a perturbation while deemphasizing common responses relies on a diverse set of active antimicrobial agents with differing MOAs in a queried batch to enable z-score standardization, we sought to assess the degree of diversity of MOAs required to still achieve accurate MOA prediction. We performed an in silico rarefaction experiment where we subset the antimicrobial reference set into a query set and a new, smaller reference set (which excluded all samples from the same batch as the query set). We constructed different query sets with different numbers of MOAs represented in them, but always including vehicle control, and used them to calculate z-scores before calculating correlations between the query compounds and the reference treatments (SI Appendix, Additional Methods). The correlations of the query compounds to their matching MOAs consistently declined and became noisier as the number of MOAs represented in the query set decreased (SI Appendix, Fig. S5). While the MOAs were predicted correctly in the tested cases, the targets were not always predicted correctly when there was only vehicle control with the query compound in the same batch (SI Appendix, Fig. S5). Thus, more diversity of MOAs represented in the queried batch generally provided stronger correlations to the reference set.

External Validation of the Antimicrobial Reference Set. To test PerSpecTM's performance on external data, we queried the largest publicly available gene expression dataset in response to antibiotic and antiseptic treatments in *P. aeruginosa* (15). Of note, the experimental design of the external query dataset was highly divergent from the methods employed in our study, most notably in their use of sublethal treatments in contrast to treatments used to generate the reference set. The external dataset was generated in duplicate for 14 antimicrobial compounds, including one fluoroquinolone (ciprofloxacin), three aminoglycosides (gentamicin, neomycin, and tobramycin), three β -lactams (ampicillin, carbenicillin, and cefoperazone), one membrane disruptor (polymyxin B), one cell wall synthesis inhibitor (aztreonam) and five antiseptics (benzalkonium chloride, bleach, peroxide, povidone-iodine, and silver nitrate). The size and chemical diversity of the query dataset allowed us to standardize gene expression with respect to the entire query dataset. SI Appendix, Table S4 summarizes the MOA and target predictions of the external chemical treatments queried against the reference set.

We applied PerSpecTM to the external transcriptional data of the 14 compounds. We could not make predictions for two compounds, aztreonam and benzylalkonium chloride, because their responses most highly correlated with negative control. Of the remaining 12 compounds, PerSpecTM correctly predicted MOAs of 10 compounds (and the targets of five compounds) (SI Appendix, Table S4). The prediction accuracy was dependent on r_{predict} values. Considering the six compounds with largest r_{predict} values (≥ 0.26), PerSpecTM

correctly predicted all six MOAs, and 4 of the 6 targets (Fig. 2A and SI Appendix, Table S4), including gentamicin, neomycin, peroxide, and ciprofloxacin. PerSpecTM predicted bleach to target DNA based on its highest similarity to nitrofurantoin, a known inducer of ROS production in bacteria. Interestingly, bleach also correlated to many of the inhibitors of membrane integrity, consistent with bleach imparting generalized oxidative damage on both DNA and membranes across diverse active treatments (50). Polymyxin's target was misassigned by a slim margin, as its third highest $\bar{r}_{\text{max}}$ corresponded to LPS, its known target (Fig. 2A).

The prediction accuracy decreased for the remaining six compounds with lower r_{predict} values, with PerSpecTM correctly predicting four MOAs and one target. Of note, all of these lower six compounds elicited weak transcriptional responses, formally fitting our defined criteria for "weak transcriptional signal" (i.e., the maximal $-\log_{10}(p_{\text{adj}}) < 20$). Thus, higher confidence predictions can be made for compounds with a higher r_{predict} value, as previously shown in the internal LOOCV analysis. Lower r_{predict} values and maximal $-\log_{10}(P\text{-values}) < 20$ tended to erode confidence in the PerSpecTM prediction.

To obtain some sense of the uncertainty in the ranking of $\bar{r}_{\text{max}}$ values, which is pertinent to the predicted target and MOA as the prediction is made based on the highest $\bar{r}_{\text{max}}$, we assessed the robustness of target predictions to fluctuations in the underlying data. We randomly subsampled 50% of the antimicrobial reference dataset and recalculated the correlation ($\bar{r}_{\text{max}}$ and r_{predict}) and thus predicted a target and MOA for the 12 compounds for which predictions could be made from the external query dataset. We repeated this 1,000 times, each time randomly selecting a different 50% of the reference set and noted the percent of times the same target was predicted as when the full reference set was used (SI Appendix, Fig. S6 and Additional Methods). In general, the target predictions were quite robust, especially for higher values of r_{predict} which correspond to higher PPVs. For example, the same target prediction of "DNA" was made for peroxide in 99% of the simulations (SI Appendix, Fig. S6). This in silico exercise thus further confirms that target predictions are more robust for higher r_{predict} values and less stable for lower r_{predict} values.

Despite the ability to predict MOA on external data by comparing z-scored expression profiles, the PPV of such predictions is lower than from the LOOCV analysis of internal reference data, as evidenced by the overall lower distributions of r_{predict} values for the external data than the internal data, as well as the lower r_{predict} values for the same compounds present in both the internal and external datasets (Fig. 2B). We attributed these differences to disparities in experimental format, including sublethal dosing and short incubation times that yielded weaker transcriptional signals. Indeed, eight out of the 14 compounds had weak transcriptional signals, where the most extreme adjusted P -value, based on analysis with DESeq2, did not exceed that of the comparison between negative control samples. In fact, compounds causing less significant gene expression changes generally corresponded to lower r_{predict} values (SI Appendix, Table S4). Overall, however, these results indicate that the standardization method was able to compensate for the technical differences between the datasets.

Finally, we compared the performance of this approach using z-scores calculated across a diverse set of query compound treatments, to using what we termed "nz-scores," which refers to standardizing gene expression relative only to negative control samples. Specifically, we compared the distribution of r_{predict} values of the external data using z-scores, calculated across the diverse sample set, versus nz-scores calculated only by comparing to negative controls. Not only was the r_{predict} distribution of the nz-score analysis much lower, but the nz-score analysis failed to make any meaningful confident predictions

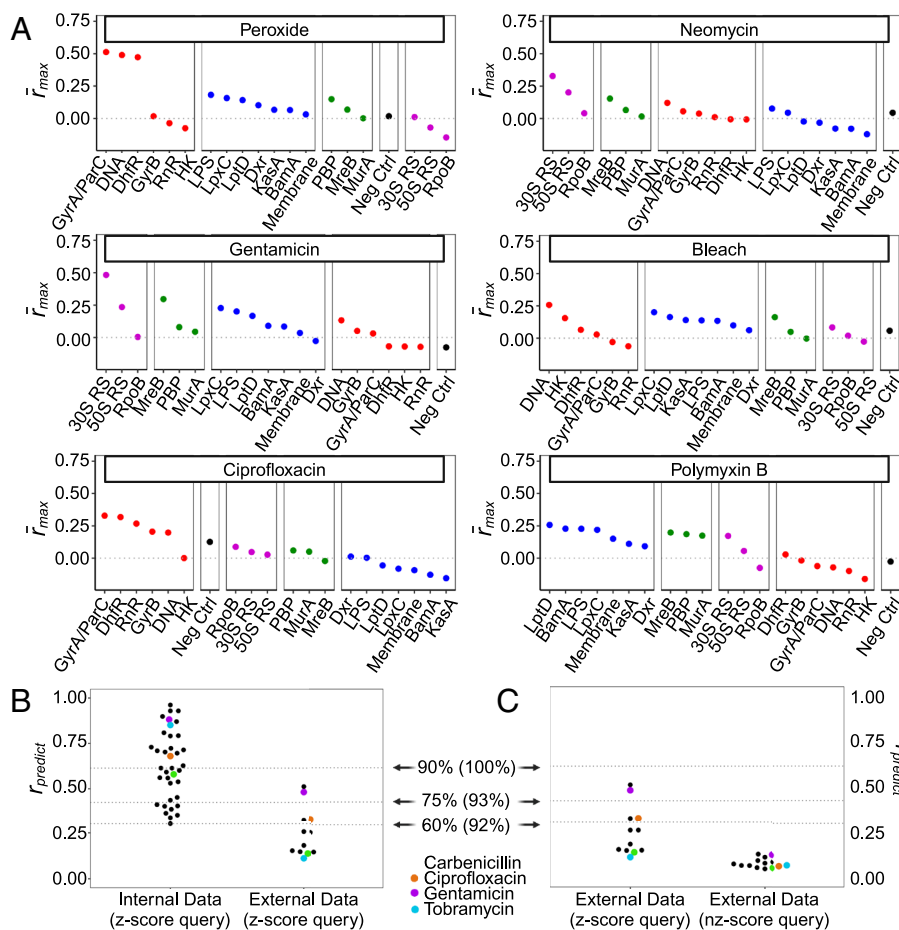


Fig. 2. PerSpecTM analysis of external dataset by mapping to the antimicrobial reference set. (A) Target correlation scores (r_{max}) of the top six external query compounds against the target categories indicated in the antimicrobial reference set. The highest r_{max} score is the target correlation prediction score ($r_{predict}$) and identifies the predicted target. Colors correspond to MOA classes: DNA synthesis (red), membrane integrity (blue), cell wall synthesis (green), and protein synthesis (magenta). (B and C) Distribution of $r_{predict}$ for all compounds in B the internal reference set in the LOOCV compared to the external query set using z-scores and (C) the external query set using z-scores compared to nz-scores. Each point represents a compound. Points of the same color identify the same compound in both the reference and external query sets, as shown. Dotted lines indicate three $r_{predict}$ thresholds, labeled with approximate target PPVs determined from the LOOCV z-score analysis (with MOA PPVs in parentheses). Target prediction scores are not indicated for compounds that were most correlated to negative control.

(Fig. 2C). Indeed, in a similar analysis, nz-scores calculated from within our internal antibiotic reference set, when used as query data, also had a lower distribution of $r_{predict}$ values and lower confidence in target predictions than when using z-scores, albeit to a lesser extent than for the external dataset (SI Appendix, Fig. S7). Of note, the external dataset only included two negative control samples, which could in part account for the poor performance using nz-scores. Taken together, these results demonstrate that by calculating z-scores across a sample set representing diverse mechanisms, PerSpecTM can yield accurate MOA and target predictions despite highly divergent methodologies.

MOA and Target Determination of Antipseudomonal Compounds Using PerSpecTM and the Antimicrobial Reference Set. To test whether PerSpecTM could predict MOAs of molecules, we focused on three small molecules we had previously identified using a multiplexed chemical genetic-based strategy termed PROSPECT (primary screening of strains to prioritize expanded chemistry and targets). PROSPECT is an antimicrobial discovery platform that identifies active small molecules based on their activity against one or a few bacterial hypomorphic mutants, each depleted of a different essential protein target (7). All three compounds were identified based on their specific activity against certain hypomorphs in which we had depleted essential outer membrane targets of PA14: PA-0918 against the *lptD*-hypomorph; PA-69180 against *oprL*-hypomorph; and PA-5750 against *lptE*-hypomorph (SI Appendix, Table S5). LptD and LptE form the LPS transport complex that shuttles LPS across the outer membrane, while OprL is a scaffolding protein involved in cell-division (51, 52). In the case of PA-0918 and PA-5750, we performed expression profiling using PA14 because both compounds were active against

the wild-type. PA-69180 had no wild-type activity, however, and therefore, we performed gene expression profiling with the *oprL*-hypomorph.

PerSpecTM predicted PA-0918 to be an inhibitor of membrane integrity with LPS as the predicted target based on an $r_{predict}$ of 0.74 (100% target PPV, 100% MOA PPV) (Fig. 3A and SI Appendix, Table S4). PA-0918 was most highly correlated with colistin, a polymyxin antibiotic known to target LPS along the outer membrane of gram-negative bacteria (53). To test whether PA-0918 shares a similar MOA with colistin as a direct membrane disruptor, we utilized an ethidium bromide uptake assay in which membrane disruption is detected by increased flux of ethidium bromide into the cells. We found that indeed, PA-0918 phenocopies colistin's ability to promote rapid uptake of ethidium bromide into cells (Fig. 3B), consistent with PA-0918 disrupting the outer membrane to increase permeability.

Using the *oprL*-hypomorph, PerSpecTM predicted PA-69180 to be a cell wall synthesis inhibitor with MreB, the major Rod protein responsible for the rod shape of gram-negative bacteria, as the predicted target based on an $r_{predict}$ of 0.38 (68% target PPV, 94% MOA PPV) (Fig. 3C and SI Appendix, Table S4). To test whether PA-69180 could be an MreB inhibitor, we used a mecillinam rescue assay described previously (54), which leverages over-expression of the cell-division protein FtsZ in *E. coli* (strain MG1655/FtsZup) to render the Rod system nonessential, thus conferring resistance to MreB inhibition. Meanwhile, MreB inhibition in this strain reverses the lethal effects of the penicillin binding protein 2 inhibitor, mecillinam, which otherwise causes a toxic malfunctioning of the Rod machinery leading to cell death. We found that PA-69180 phenocopies the known MreB inhibitor A22, with MG1655/FtsZup being both resistant to PA-69180 killing

and resistant to mecillinam killing upon the addition of PA-69180 (Fig. 3D), which was not the case for the negative control ciprofloxacin (SI Appendix, Fig. S8). We also generated resistant mutants to PA-69180 in the efflux-deficient strain PAO397 and performed whole-genome sequencing to reveal two mutants, PAO397-r1 and PAO397-r2, revealing mutations at distinct amino acid positions adjacent to the ATP binding pocket of MreB (55) that conferred high-level resistance to PA-69180 and cross-resistance to A22, thereby confirming MreB as the target (SI Appendix, Table S6).

Finally, PA-5750 was assigned as a DNA synthesis inhibitor with GyrA as the predicted target based on an r_{predict} value of 0.56 (88% target PPV, 96% MOA PPV) (Fig. 3E and SI Appendix, Table S4). Based on this prediction, we moved to immediately test PA-5750's ability to both intercalate DNA and inhibit DNA gyrase. We found no evidence of DNA intercalation by PA-5750 based on its inability to stimulate supercoiling activity of topoisomerase I or to displace ethidium bromide from purified plasmid, in contrast to the known DNA intercalator m-amsacrine (SI Appendix, Fig. S9 A and B). Instead, we found that PA-5750 inhibited *E. coli* DNA gyrase's ability to supercoil DNA while the inactive enantiomer of PA-5750 failed to do so (SI Appendix, Fig. S9C). Finally, using two ciprofloxacin-resistant mutants of *E. coli* J53 strain, which each harbor different GyrA point mutations, we demonstrated cross-resistance to PA-5750 (Fig. 3F), but not to the GyrB inhibitor novobiocin (SI Appendix, Fig. S10). Taken together, these results support PerSpecTM's prediction that PA-5750 targets gyrase, and overall, validate its ability to predict MOA for novel compounds.

Correlating the Transcriptional Responses of Hypomorphic Strains. While we demonstrated the ability of PerSpecTM to predict small-molecule MOAs using an antimicrobial reference set, the limited number of known inhibitors and their small set of targets and MOAs could restrict the utility of such an approach, particularly with respect to assigning MOA to compounds with completely novel targets and MOAs. We thus sought to test whether the transcriptional responses of hypomorphic strains, in which essential targets have been depleted, correlate with chemical inhibition of the corresponding targets. If so, PerSpecTM could be used in a reference-based manner to correlate responses between small-molecule inhibition to genetic depletion of the same target, thereby providing a comprehensive map of all targetable pathways.

We initially analyzed expression data from the seven hypomorphs we had previously engineered using a regulated promoter replacement strategy to deplete essential gene targets. However, the transcriptional changes in these strains relative to wild-type bacteria were minimal. Based on our findings with small molecules, we wondered whether the transcriptional changes in these strains relative to wild-type bacteria could be minimal due to insufficient strength of knockdown provided by the promoter replacement system. We therefore shifted strategies to use CRISPRi technology to create an inducible gene repression system. We constructed 19 CRISPRi strains encoding arabinose-induced dCas9 and constitutively expressing guide RNAs targeting *dhfR*, *drr*, *gyrA/gyrA-2* (two different strains with different guides), *gyrB*, *leuS*, *lptA*, *lptD*, *lptE*, *lpxC*, *mreB*, *murA*, *parC*, *pbpA*, *ssb*, *rplJ*, *rpoB*, *rpsL*, and a no gene control (referred to as CRISPRi-C26),

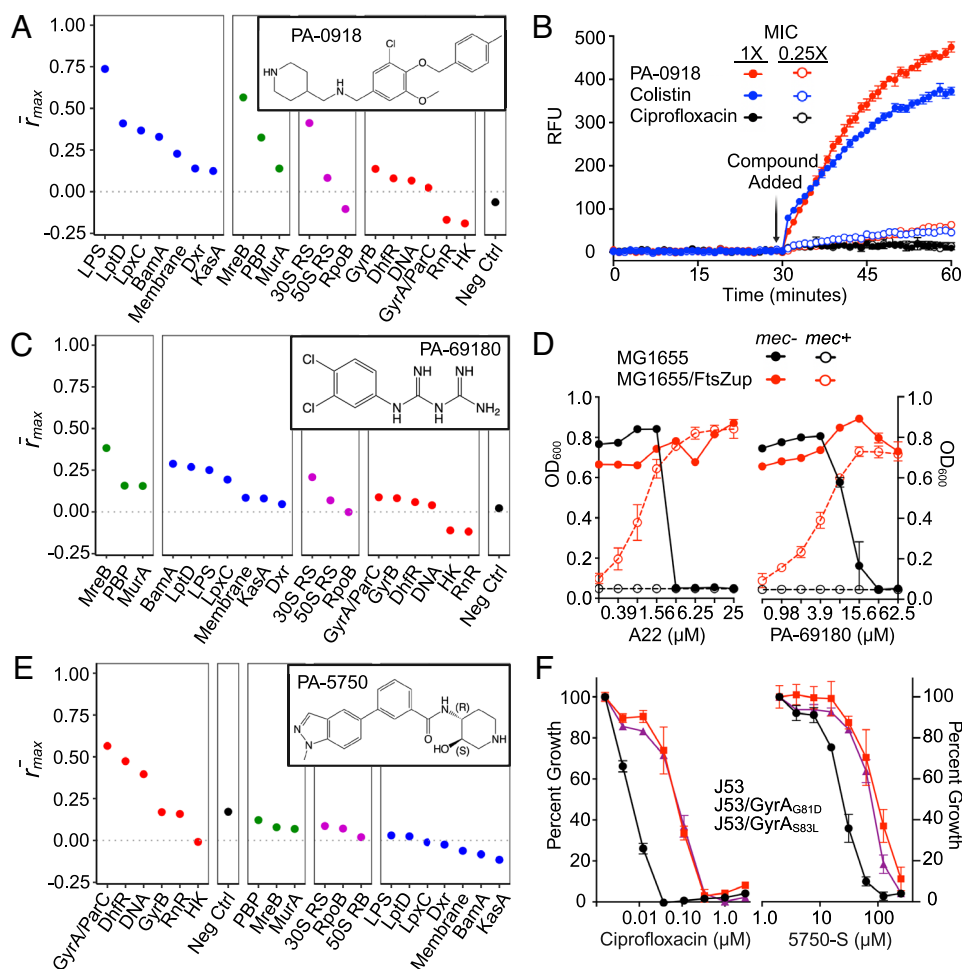


Fig. 3. MOA determination of antipseudomonal compounds. (A) Target correlation scores (r_{max}) of PA14 treated with PA-0918 against the target categories indicated in the antimicrobial reference set. PA-0918 is predicted to be a direct LPS binder, similar to colistin, based on a target correlation prediction score (r_{predict}) of 0.74 (target PPV of 100%). (B) Ethidium bromide uptake in cells treated with PA-0918 and colistin, with ciprofloxacin as a negative control. PA-0918 treatment phenocopies colistin with rapid uptake of ethidium bromide suggestive of membrane disruption. In all experiments, error bars represent SEM of three biological replicates ($n = 3$). (C) r_{max} scores of *opRL*-hypomorph treated with PA-69180 against the target categories indicated in the antimicrobial reference set. PA-69180 is predicted to be an MreB inhibitor, similar to A22, based on an r_{predict} of 0.38 (target PPV of 68%). (D) MG1655 overexpressing *ftsZ* (MG1655/ FtsZup) is resistant to both A22 and PA-69180 and is rescued from mecillinam killing (*mec*- and *mec*+ denote 0 $\mu\text{g/mL}$ and 2.5 $\mu\text{g/mL}$ mecillinam, respectively) by both compounds, implicating MreB as the target. (E) r_{max} scores of PA14 treated with PA-5750 against the target categories indicated in the antimicrobial reference set. PA-5750 is predicted to be a GyrA/ParC inhibitor, similar to ciprofloxacin, based on a r_{predict} of 0.56 (target PPV 88%). (F) Two ciprofloxacin-resistant *E. coli* J53 mutant strains, each containing single point mutations (G81D or S83L) in GyrA, are cross-resistant to PA-5750, implicating DNA gyrase at the target.

summarized in *SI Appendix, Table S7* and *Additional Methods* for details of strain construction). In each CRISPRi strain, the addition of arabinose-induced dCas9 expression and subsequent transcriptional repression of the target gene of interest. We characterized the arabinose-dependent growth phenotypes of each CRISPRi strain (*SI Appendix, Fig. S11*). The CRISPRi-C26 control strain showed no significant growth delay over an arabinose range of 0 to 0.5% (w/v), indicating no appreciable dCas9-mediated toxicity in PA14. Mutants targeting *dhfr*, *lptA*, *lpxC*, *mreB*, *murA*, and *parC* all demonstrated severe arabinose dose-dependent growth delays, whereas mutants targeting *dxr*, *gyrA*, *gyrB*, *leuS*, *lptD*, *lptE*, *rplJ*, *rpoB*, and *rpsL* exhibited more moderate growth delays at higher arabinose concentrations. By comparison, mutants targeting *pbpA* and *ssb* showed no growth delay.

We performed transcriptional profiling of all 19 CRISPRi strains in 384-well format at 360 min in 0%, 0.125%, and 0.5% arabinose before chemical lysis, RNA extraction, and RNAseq library construction. Z-scores were calculated across all strains and arabinose doses, thereby accentuating strain-specific transcriptional changes. Additionally, we used DESeq2 to assess the level of knock-down of the targeted genes in each condition, as summarized in *SI Appendix, Table S7*. In general, CRISPRi strains demonstrated an arabinose-dependent decrease in the expected target mRNA levels compared to levels in the CRISPRi-C26 control strain, ranging from 1.5 to 174-fold in 0.125% or 0.5% arabinose for conditions yielding significant on-target gene expression changes (adjusted *P*-value < 0.05). Of note, no statistically significant expression changes in the target gene were detected even at the highest arabinose concentrations for mutants carrying guides targeting *lptE*, *ssb*, and *rpoB*, indicating that appropriate, effective guides were not utilized for these genes.

To form a genetic depletion reference set using robust transcriptional profiles from the CRISPRi strains, we compiled all treatment conditions with greater than 0% arabinose. We again removed treatment conditions yielding transcriptionally weak responses, defined as the differential expression between arabinose treatment and control guide C26 wherein no gene had an adjusted *P*-value that was more significant than the minimum adjusted *P*-value defined previously by comparing PA14 in water versus DMSO control (i.e., $p_{\text{adj}} < 10^{-20}$). Applying this filter, mutants carrying guides targeting *dhfr*, *leuS*, *lptE*, and *gyrA-2* (i.e., the CRISPRi-*gyrA-2* strain) were removed from analysis, leaving 14 strains in the genetic depletion reference set, with some comprising multiple levels of knockdown induced by different arabinose doses, in addition to the control strain.

Finally, we applied PerSpecTM using the genetic depletion reference set to assess whether the transcriptional profiles of the CRISPRi strains could serve as a reference. The target correlation scores $\bar{r}_{\text{max}}$ were calculated between the z-scored gene expression profiles of CRISPRi strains (as the reference set) and those of compound treatments (the query samples) of PA14, except for fosmidomycin, which was tested in the *dxr*-hypomorph strain due to its inactivity against PA14. As shown in Fig. 4, the transcriptional responses of antibiotic treatments generally correlated to transcriptional responses of the corresponding CRISPRi strains depleted for the cognate gene targets, or their associated ribosomal/protein complexes. The target was predicted correctly in some cases. Ciprofloxacin treatment correlated strongest with *parC* depletion, followed by *gyrA*, accurately mapping to its known targets of GyrA and ParC. Similarly, the GyrB inhibitor novobiocin correlated most strongly to depletion of *gyrB*, the LpxC inhibitor PF-04753299 mapped to *lpxC* depletion, the LPS transport inhibitor POL7080 mapped to *lptA* depletion, the MurA inhibitor fosfomycin mapped to *murA* depletion, and the

aminoglycoside amikacin mapped to ribosomal subunit *rpsL* depletion. Of note, strains depleted for *parC* (a topoisomerase) and *lptA* (part of the LPS transport complex), which showed some of the greatest fitness defects, yielded the strongest r_{predict} values of 0.56 and 0.46 upon inhibition with ciprofloxacin and POL7080, respectively. When the target was not correctly predicted, in most cases PerSpecTM accurately predicted the MOAs. For example, all protein synthesis inhibitors (amikacin, clarithromycin, and rifampicin) correlated most strongly with the two CRISPRi strains depleted of targets which interfere with protein synthesis (*rpsL*, *rplJ*, and *rpoB*). Similarly, fosmidomycin, which was misassigned as an LptA inhibitor, was correctly predicted to disrupt membrane integrity. Indeed, the role of its target Dxr in the synthesis of isoprenoid precursors is required for maintenance of the cell wall and thus membrane integrity. Of the two compounds with misassigned MOAs altogether, the cell wall synthesis inhibitors A22 and imipenem, both were predicted to be membrane disruptors while correlating to a lesser extent with CRISPRi strains disrupting cell wall synthesis. In general, like the small-molecule reference set, there is significant overlap in transcriptional changes between inhibitors of membrane integrity and cell wall synthesis with compounds tending to correlate with gene depletions across both mechanistic classes, suggesting a shared transcriptional response to disruptors of both cellular processes. Together, these data demonstrate that the transcriptional response of chemical inhibition of a given target correlates with that of CRISPRi-induced repression of the same target.

Discussion

Target identification is central to any antibiotic discovery pipeline to guide candidate prioritization and development. The advancement of more sophisticated approaches to antibiotic discovery is working to expand the chemical space and increase the number of primary candidates discovered with antibacterial activity. Meanwhile, these novel platforms highlight the bottleneck that target identification imposes on antibiotic development efforts. While measuring the transcriptional response to small-molecule perturbation has now become quite a standard step in characterizing a new antibiotic candidate, its value in MOA elucidation is often supportive at best, despite the anticipation that it could yield valuable insights into the biology of how target disruption leads to killing, thus leading to a MOA. To date, the interpretation of transcriptional responses has been confounded by background changes in gene expression that are secondary or even unrelated to the immediate, direct effect of target inhibition—such as those relating to growth, general stress, or cell death. These changes can drown out perturbation-specific gene responses. The major approach that has been taken to mitigate these confounders is to tailor experimental designs to control for background gene expression by treating with subinhibitory concentrations while carefully trying to match growth rates; these approaches, however, limit the strength of the elicited transcriptional responses. An alternative approach has been to develop computational algorithms that select the most informative genes, which in some implementations may be at risk of overfitting the reference dataset if the selected genes are optimized based on their ability to predict the MOAs of the compounds in the reference set.

Here, we developed a unique strategy, PerSpecTM, that works by leveraging the ability to scale the measurement of expression profiles across a large set of active, diverse perturbations, thereby enabling a standardization (calculation of z-scores) that deemphasizes the background changes to highlight perturbation-specific changes as the basis for more accurate target and MOA prediction. We demonstrate proof of principle that by applying PerSpecTM's

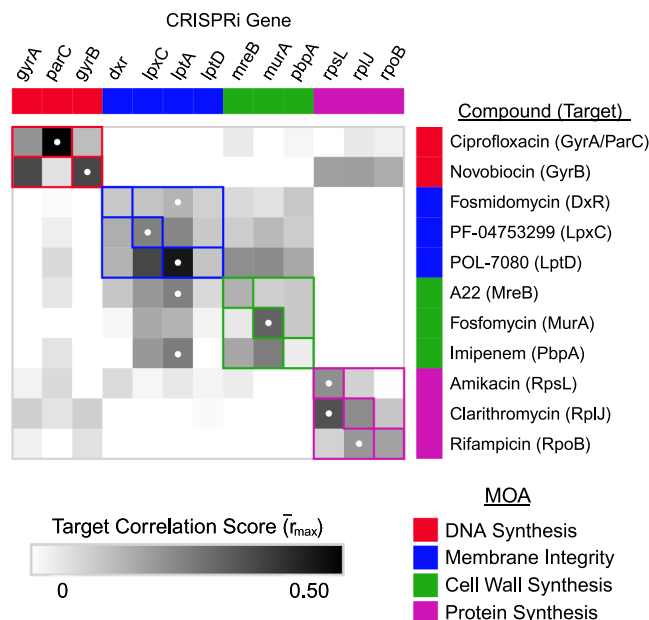


Fig. 4. PerSpecTM analysis of antimicrobial treatments queried against the genetic depletion reference set. Heatmap intensities show target correlation scores ($\bar{r}_{max}$) of 90-min antibiotic treatments when queried against CRISPRi strains in the genetic depletion reference set treated with 0.125% or 0.5% arabinose to induce dCas9 expression. PA14 was treated with all query compounds except fosmidomycin, which lacked wild-type activity, and thus the *dxr*-hypomorph was used. White dots indicate the target correlation prediction score (r_{pred}) for each query compound. Colored boxes indicate the MOA relating to each target gene or query compound. Note that from the genetic depletion reference set, only those that have corresponding compound treatments targeting the same gene targets, or the associated ribosomal/protein complexes, are shown.

reference-based approach, either to a set of known antimicrobial compounds with known MOA or a set of strains which are genetically engineered to be depleted of different essential targets, a compound's MOA and even target can be reasonably predicted. We validated PerSpecTM's performance on both internal and external small-molecule datasets relative to an antimicrobial reference set, as well as proof of principle using a collection of 14 CRISPRi depletion strains.

Importantly, we found that eliciting a strong transcriptional response was critical to PerSpecTM's ability to predict MOA in the case of both chemical and genetic perturbations. Reasonably potent antimicrobial activity (i.e., being able to treat with above-MIC concentrations) was necessary to ensure a robust transcriptional response to chemical treatments; this however poses a barrier for early MOA prediction of new compounds that have not yet been optimized to have potent wild-type activity. For weakly active molecules, their solubility limits could prevent treatment of wild-type bacteria at twofold to fourfold above the MIC. Meanwhile, such weak inhibitors could be of significant value either as early antibiotic candidates or as chemical biological probes, particularly if they work by a novel MOA. In such cases, we show that hypersusceptible strains can be substituted for wild-type bacteria, with the changes in such strains mapping to the same changes in wild-type bacteria, dose adjusted. Thus, engineered, hypersensitive hypomorphs can enable PerSpecTM's ability to predict MOA in the case of weak inhibitors. Meanwhile, we also found that not all genetically engineered hypomorphs have transcriptional profiles that differ significantly from those of wild-type bacteria. Sufficiently strong knockdown, for example in the engineered CRISPRi strains versus the promoter replacement strains, is critical to eliciting a robust transcriptional

difference between mutant and wild-type bacteria, even while the level of knockdown required to generate these differing responses will vary from gene to gene. This is supported by the fact that the least fit CRISPRi strains – with growth impairment reflecting sufficient target depletion to impact cellular physiology—yielded the strongest Pearson correlations to chemical inhibition of their corresponding gene targets. Thus, the more impaired a strain's fitness is, the more informative its transcriptional profile can be.

We applied PerSpecTM to predict MOA for three active antipseudomonal compounds that we had identified—PA-0918, PA-69180, and PA-5750—to highlight its power in MOA determination early in the antibiotic discovery process. All three compounds originated directly from chemical libraries and therefore had not yet undergone extensive structure–activity relationship studies or optimization. PerSpecTM predicted PA-0918 to be a membrane disruptor, PA-69180 to be an MreB inhibitor, and PA-5750 to be an inhibitor of DNA gyrase, all of which we subsequently confirmed with supporting experimental validation. The case of PA-69180, which lacked significant wild-type activity, highlights the advantage of leveraging the hypersusceptibility of the *oprL*-hypomorph to generate a robust transcriptional signal, which enabled a correct, high-confidence MOA prediction. Interestingly, it also suggests a potential functional interaction between OprL and MreB, relating cell shape and structure to cell wall integrity. The case of PA-5750 adeptly highlights the advantage of integrating MOA prediction tools early in the discovery process, as PerSpecTM's prediction immediately focused our efforts on the hypothesis that PA-5750 inhibits DNA synthesis. Taken together, these data lend credence to the ability of transcriptional profiling to expedite the incorporation of MOA prediction into antibiotic discovery processes, aiding in prioritization before any chemical optimization.

Referenced-based MOA prediction by expression profile mapping to a collection of known antimicrobial compounds is inherently limited by the diversity of the reference set, which poses a barrier to discovering inhibitors of novel biological pathways without known chemical inhibitors to anchor predictions. We therefore explored a complementary approach of mapping to genetic perturbations by constructing a transcriptionally repressed reference set using CRISPRi technology. While chemical inhibition of a series of gene targets correlated strongest with their corresponding CRISPRi knockdown strains and the best correlations were made when the level of knockdown was strong enough to impair fitness, some strains without impaired fitness nevertheless provided informative expression changes. In future iterations of this approach, generating strains in which bacterial fitness is compromised as part of a larger, more comprehensive library of CRISPRi strains, potentially guided by genome-wide fitness studies as have been reported for *M. tuberculosis* (56), would yield a stronger genetic perturbation reference set from which one can generate more robust correlations for MOA prediction. Nevertheless, these data provide proof of concept that a dedicated genetic perturbation reference set could facilitate accurate MOA prediction, including for gene targets without known chemical inhibitors. In theory, a comprehensive library of CRISPRi strains could be used to define the transcriptional map of all essential genes of a certain bacterial species, serving as a reference set for completely unbiased MOA predictions.

Nevertheless, several caveats in the PerSpecTM approach warrant comment. PerSpecTM remains a reference-based strategy; therefore, by relying on an antimicrobial reference set as the basis for MOA prediction, PerSpecTM will inherently be limited to predicting MOA for novel compounds which work by known, established mechanisms or targets for which there are preexisting inhibitors. This limitation can be circumvented by constructing

reference sets of hypomorphic mutants, based on the assumption that gene depletion phenocopies target inhibition. While this assumption is often true, it is not universally so. For example, differences in the timescale of loss of function between small-molecule inhibition and gene depletion can yield different transcriptional responses as cells adapt with differing compensatory mechanisms. Alternatively, complete absence of a protein may not phenocopy inhibition of a particular protein function, particularly when a protein target has multiple functions.

Relying on a genetic depletion reference set as the basis for MOA prediction also has limitations. First, as not all antibacterial compounds work via a protein target depletion (e.g., membrane disruptors, DNA intercalators, free radical producers), and transcriptional response of depleted essential protein targets may not always phenocopy the response from queried compounds. Second, the specific cellular response to some small-molecule or genetic perturbations may not be well reflected in strong, canonical transcriptional responses, instead resulting in other changes (e.g., metabolic or structural changes that lead to death). β -lactams are such a case where fewer changes in gene expression occur on a relatively slow timescale compared to other antibiotic classes. Finally, whether using an antimicrobial or genetic depletion reference dataset, the MOA prediction resolution of PerSpecTM can be compromised by the complexity of cellular networks which have significant interactions and cross talk, as evidenced by the overlapping transcriptional profiles of disruption of the cell wall and membrane integrity; inhibition of targets which share common functions or that are highly interactive can generate similar, overlapping transcriptional responses. Meanwhile, off-target effects of small molecules, particularly of weakly potent ones, can similarly generate complex, noisy transcriptional programs that are uninterpretable and do not map to inhibitors of single targets or single gene depletions.

At the same time, we note an important caveat to PerSpecTM's z-score-based approach. The power of the z-score metric is both in its simplicity as well as its ability to emphasize the informative, relevant signal while deemphasizing the noise that is common to all death-inducing perturbations. This work demonstrates the reasonably good performance of such an approach. However, the ability to statistically remove the common noise without losing informative signal in this manner relies on the unbiased representation of a balanced, diverse, distinct set of perturbations analyzed within a batch. Biased MOA representation within the batch might allow a single MOA to dominate, resulting in a z-score analysis dampening signals that are common to, but also potentially meaningful across, a large number of samples. In the rarefaction analysis we performed on the internal reference set, we found that at a minimum, one reference compound with a differing MOA, as well as one negative control, is required in a batch with a query compound to yield a highly accurate prediction. However, correlations of the query compounds to their matching MOAs consistently declined and became noisier as the number of MOAs represented in the query set decreased; thus, having more compounds with greater MOA diversity within a query set will yield stronger correlations.

In this work, we tried to minimize bias by ensuring that minimally, each queried batch included control perturbations resulting from inhibition of DNA synthesis, protein synthesis, and either cell wall or membrane integrity (or both), along with negative control samples. With this representation, PerSpecTM performed well for these datasets, including an external dataset with slightly different MOA composition than the antimicrobial reference set, particularly for conditions that exhibited a robust transcriptional response. Further work would be needed to determine whether the set of control inhibitors we used are sufficiently balanced and unbiased, such that analysis of data using a different albeit still diverse set of controls in a new query set would replicate this MOA

prediction accuracy. Finally, further effort to optimize standardization with respect to negative controls (nz-scores) would be needed to enable accurate prediction as an alternative application of PerSpecTM, without the need for balanced representation of diverse control inhibitors in each batch.

Here, we present a strategy for increasing the ability for transcription-based mapping to predict MOA as well as to increase confidence in predictions with the goal of elevating gene expression profiling to be a driver for primary MOA prediction rather than just as an approach to provide ancillary data supportive of an MOA. As such, in a reference-based manner, the expression response to a small-molecule perturbation can serve as a phenotype and can be used in conjunction with other phenotypic measurements, including microscopy (57–59), chemical–genetic interaction profiles (7, 60, 61), and phenomic profiling using gene deletion libraries or transposon libraries for network mapping of gene interactions in response to chemical treatments for MOA prediction (15, 62, 63). As we look to the future, the development of a comprehensive suite of complementary phenotypic measurements resulting from small-molecule perturbation will hopefully enable more rapid and efficient MOA elucidation, thereby accelerating the development of antibiotic candidates to address the growing problem of antibiotic resistance.

Materials and Methods

Bacterial Strain Construction. *P. aeruginosa* strains were derived from the UCBPP-PA14 lab strain described previously (64). [SI Appendix, Tables S8–S10](#) describe all plasmids, strains, and primers used in this study. Detailed methods of strain construction are available in the [SI Appendix, Additional Methods](#).

Gene Expression Experiments and Library Construction. Gene expression experiments were carried out in biological triplicate in 384-well microplates (Thermo Fisher Scientific, Waltham MA). Stock solutions of compounds were dissolved fresh in DMSO or water, and 30 μ L of compound solutions were mixed with 30 μ L of mid-log culture to yield 1×10^8 CFU/mL bacteria and compounds in 0.5% (v/v) water (or DMSO) in Lysogeny Broth. Cells were treated at 37 °C without shaking in a humidity chamber for 30, 60, 90, and 120 min. In the case of the CRISPRi strains, cells were treated for 360 min. At the end of the incubation period, samples were then mixed with 30 μ L of 3 $\times$ RNAgem Blue Buffer (Custom Science, New Zealand) and chemically lysed by incubation in a thermocycler at 75 °C for 10 min. Total RNA was extracted using the Direct-zol kit (Zymo Research, Irvine CA), and RNA quality and quantity were analyzed using the RNA ScreenTape with the 2200 TapeStation (Agilent, Santa Clara CA). Illumina complementary DNA (cDNA) libraries were generated using a modified version of the RNAtag-seq protocol (65), as outlined in more detail in the [SI Appendix, Additional Methods](#).

Illumina cDNA libraries were generated using a modified version of the RNAtag-seq protocol (65). 250 ng of extracted RNA was fragmented, dephosphorylated, and ligated to DNA adapters carrying 5'- AN_8 -3' barcodes of known sequence with a 5' phosphate and a 3' blocking group. Barcoded RNAs were pooled and depleted of rRNA using the Pan-Bacteria riboPOOL depletion kit (siTOOLs Biotech, Galen Laboratories). Pools of barcoded RNAs were converted to Illumina cDNA libraries in two main steps: i) RT of the RNA using a primer designed to the constant region of the barcoded adaptor with addition of an adapter to the 3' end of the cDNA by template switching using SMARTScribe (Clontech) as described (66) and ii) PCR amplification using primers whose 5' ends target the constant regions of the 3' or 5' adaptors and whose 3' ends contain the full Illumina P5 or P7 sequences. cDNA libraries were sequenced on the Illumina (NovaSeq SP 100) platform to generate paired-end reads.

Computational Analyses, Scripts, and Data Accessibility. Computational analyses of gene expression profiling are detailed in the [SI Appendix, Additional Methods](#). Scripts for calculating z-scores, nz-scores, and predicting compound targets were written in R and stored in GitHub (https://github.com/broadinstitute/psa_rnaseq_manuscript_rproject). Raw sequencing data and z-score values can be found on the Gene Expression Omnibus (GEO accession GSE251671) (67).

Biochemical Characterization of Screening Hits. Growth kinetics, MIC experiments, resistance selection, as well as assays quantifying ethidium bromide uptake and displacement, mecillinam rescue, DNA intercalation, and DNA gyrase activity, are described in detail in the *SI Appendix, Additional Methods*.

Data, Materials, and Software Availability. Gene expression data have been deposited in GEO Database ([GSE251671](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE251671)) (67). All study data are included in the article and/or [supporting information](#).

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